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OPEN ASIA UNIFYING SCIENCE, EMPOWERING INNOVATION

Co-funded by the European Union

Bridging the Gap: Open Science Interoperability and FAIR Data in India

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Dean

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SVKM's NMIMS (Deemed to-be) University,

Mumbai, Maharashtra, India

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NMIMS at a Glance

44+
Years of Legacy

35000+
Full-time students

950+
Faculty Members

8
Campuses

17
Specialized Schools

ACCREDITATIONS

AACSB ACCREDITED
SBM - Mumbai Campus

AACSB

SBM has been **re-accredited by AACSB**, making it the first B-School in Western India to achieve this honour.

EQUIS ACCREDITED

EQUIS

NMIMS Deemed-to-be University **Accredited by EQUIS** (Mumbai Campus)

GRADE A++
Accredited by NAAC

NAAC A++

NMIMS Deemed-to-be University Accredited by NAAC with **3.67 CGPA in 2025** (Mumbai Campus)



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Why open science matters for India's Research and Innovation ecosystem.


INDIA'S RESEARCH DIVERSITY:
INNOVATION | TRADITION | PROGRESS
Space Exploration
Biotechnology & Medicine
Agriculture & Sustainability
Information Technology & AI
Traditional Knowledge & Ayurveda
Nuclear Physics & Engineering
Space Exploration
Dr. Priya Sharma (40s)
Biotechnology & Medicine
Vaccine development
Agriculture & Sustainability
Dr. Rajesh Kumar, Geneticlily
(pardy)
Information Technology
Aisha Khan, 30s
Traditional Knowledge & Ayurveda
Dr. Suresh Iyer
Nuclear Physics & Engineering
Enginers (30s, 30s

Challenges to Open Science

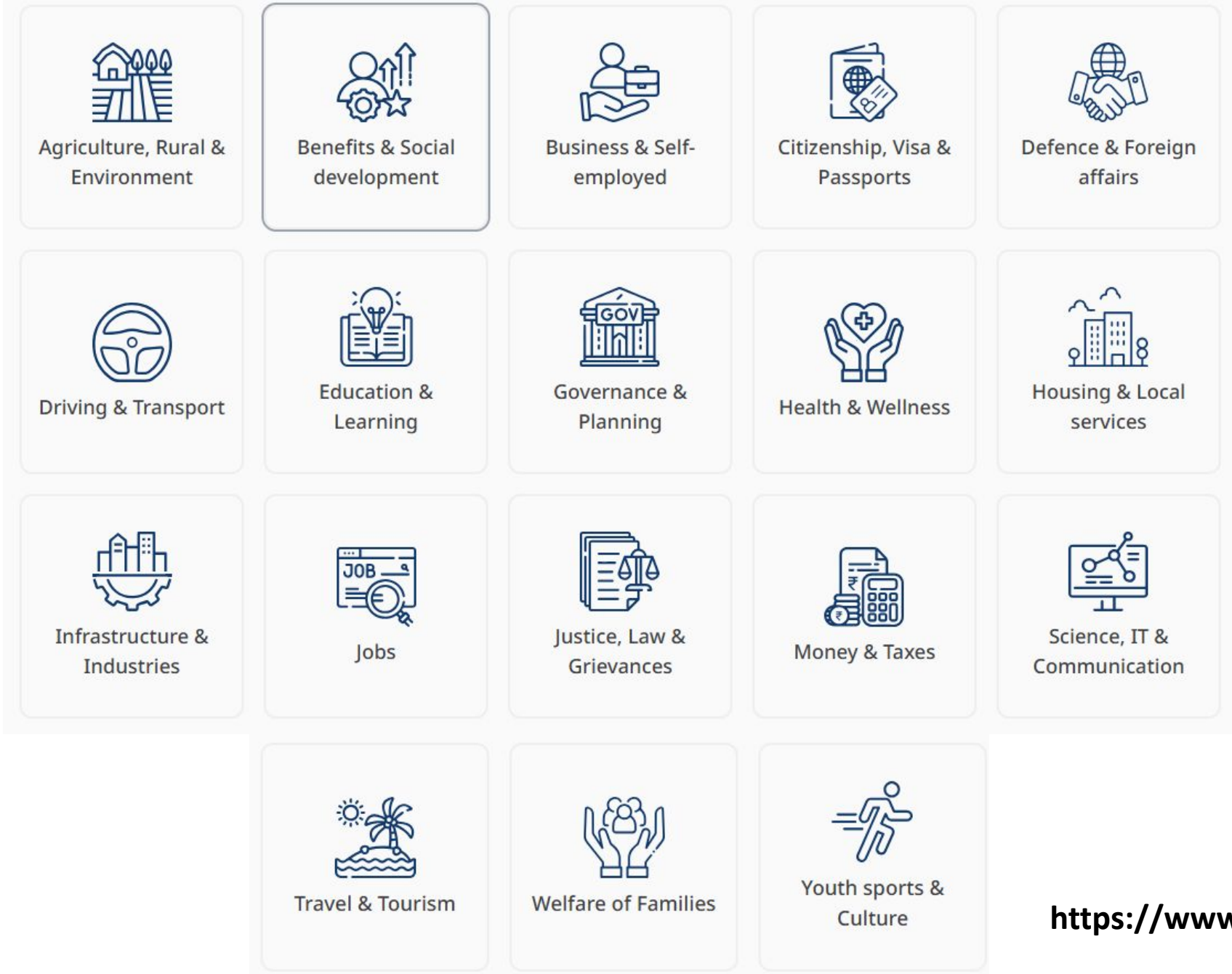
- **National policy**

National Science, Technology and Innovation Policy (STIP) 2020 remained a draft document.

- **Volume of data and Scattered data**
- **Different file formats, identifiers and lack of policies create barriers.**
- **Data is often difficult to discover, interpret, combine, and reuse.**
- **Limited training, infrastructure, funding, and incentives.**

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Services by Category

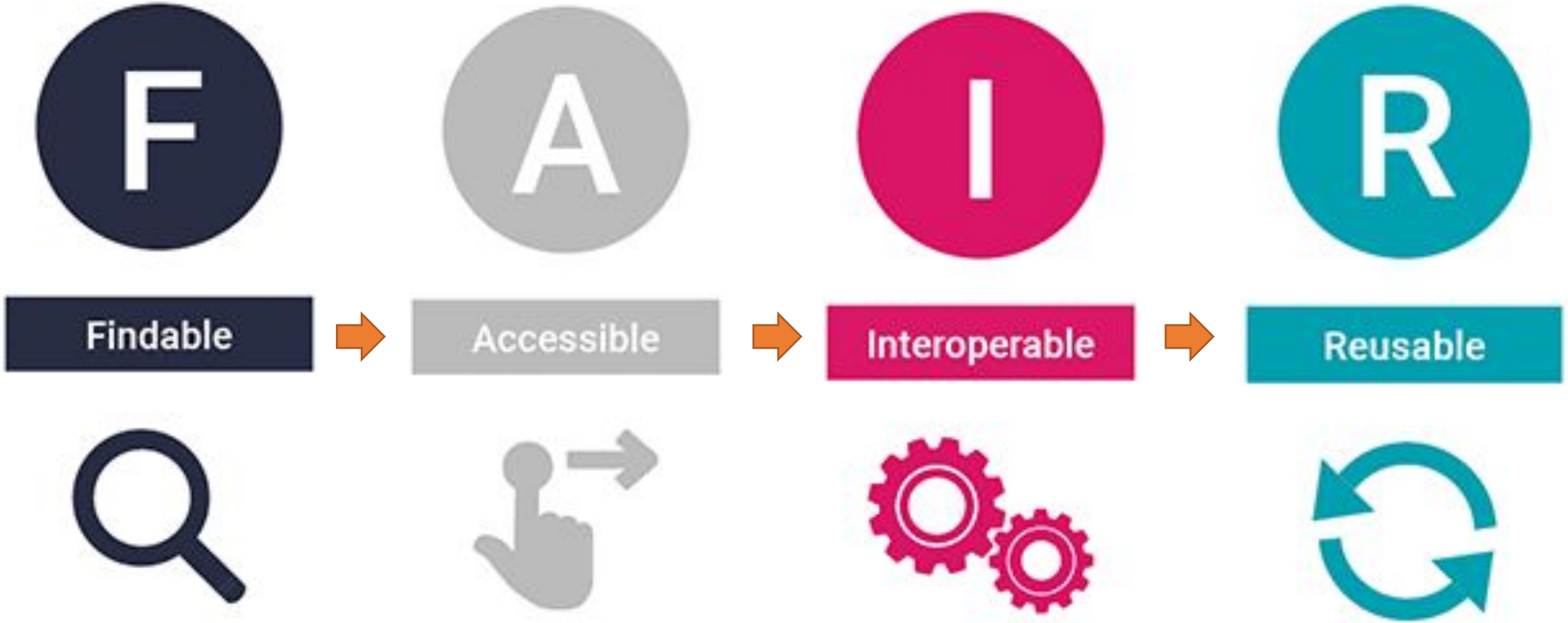


https://www.india.gov.in/services

Barriers to Open Science

- **Lack of awareness:** Researchers may lack knowledge of open science practices and how to implement them.
- **Limited resources:** Time and financial constraints may hinder adoption.
- **Fear of misuse:** Researchers may worry about competitors benefiting from shared work or data being misused or misinterpreted.

What is FAIR Data?



Building a system where data is **easy to** Search, Access, Share, and Reuse

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Importance of FAIR System



Data Silos



Lack of Discoverability

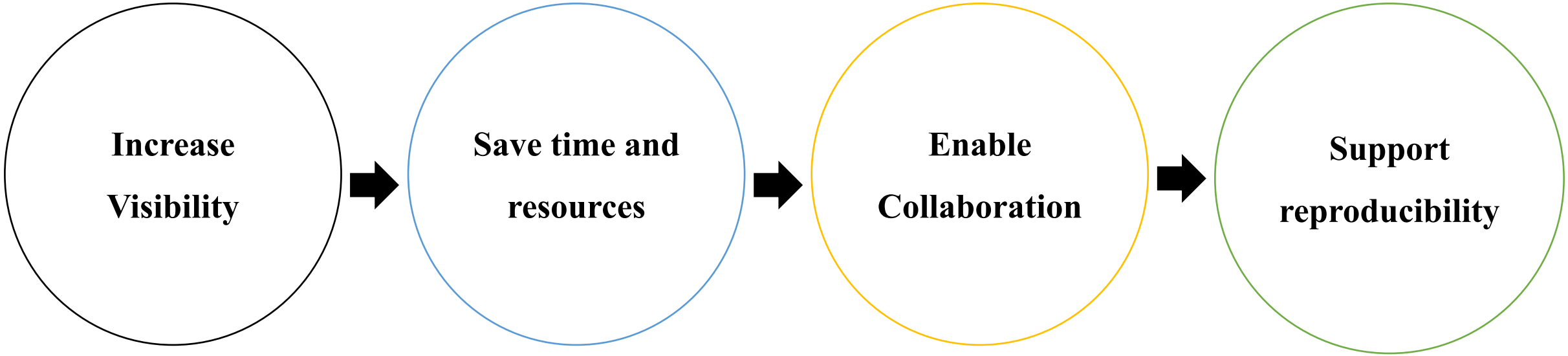


Poor MetaData Quality

Importance of FAIR System

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FAIR Principles - Ensuring data is usable by humans and machines.



FAIR Principles

Principal 1: Findable

- Persistent Identifiers e.g. DOI
- Easily available metadata e.g. Zenodo

Principal 2: Accessible

- Standard protocols e.g. HTTP
- Metadata available

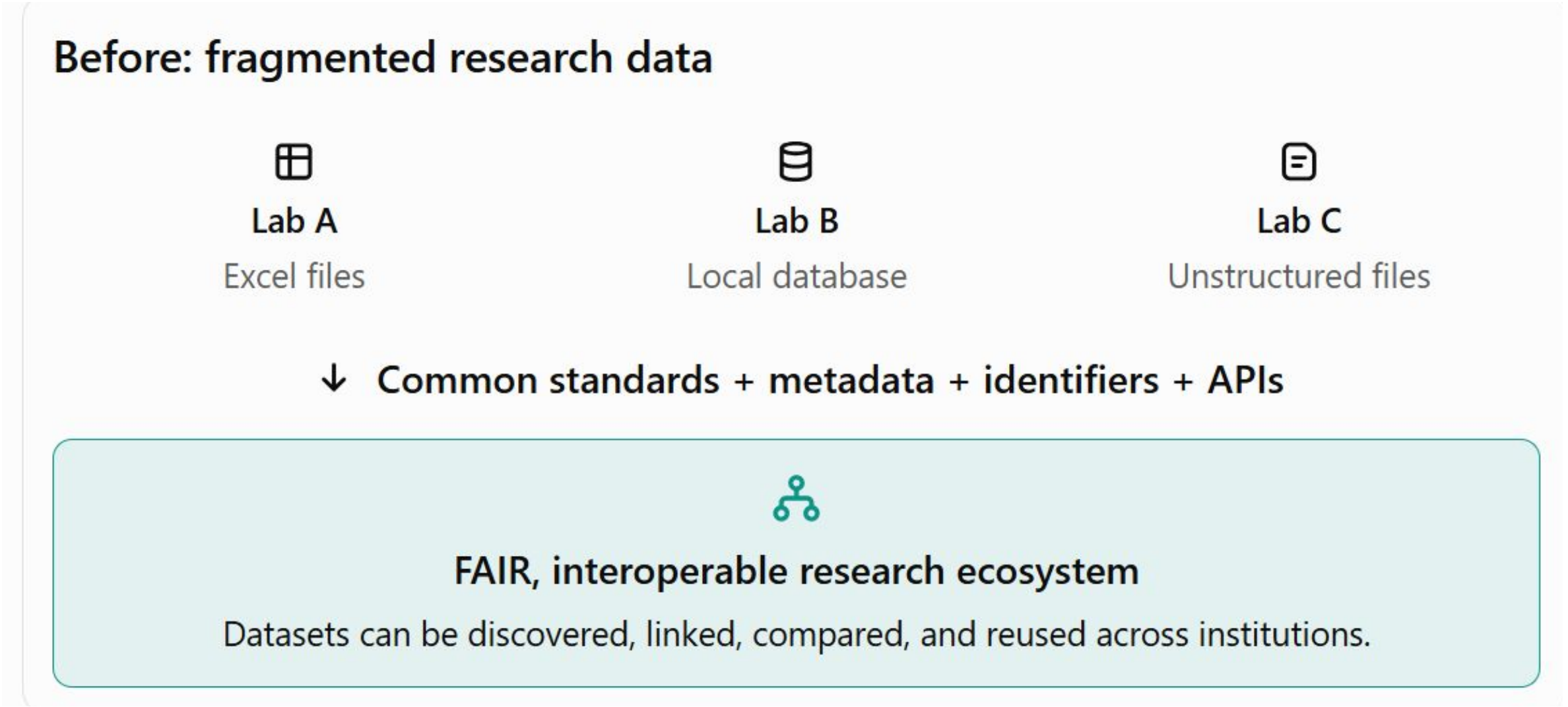
Principal 3: Interoperable

- Standard data format (CSV, JSON etc)
- Linking to related datasets & external repositories

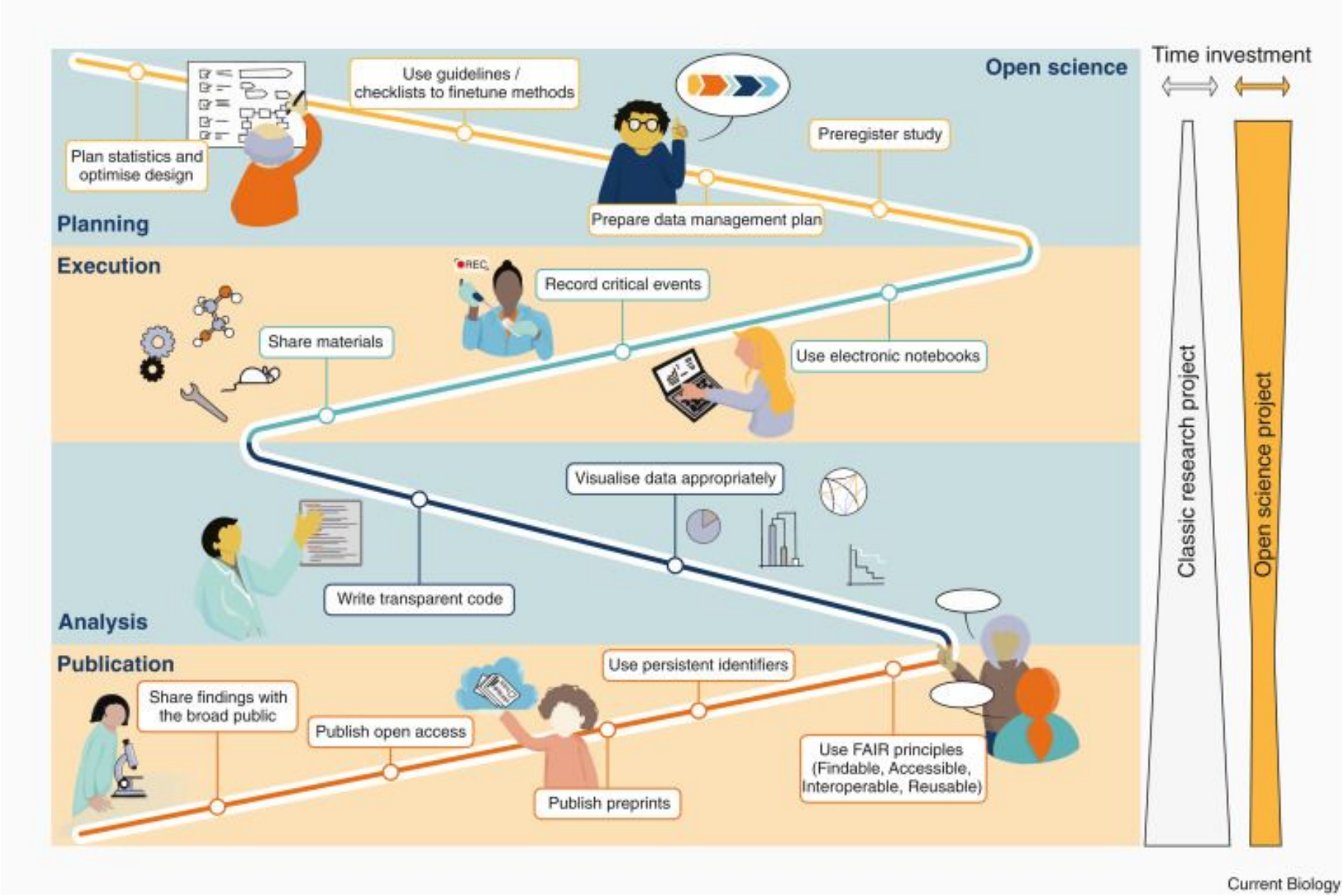
Principal 4: Accessible

- Clear licenses (CC-BY, CC0)
- Community-agreed metadata standards

What does interoperability look like in practice?

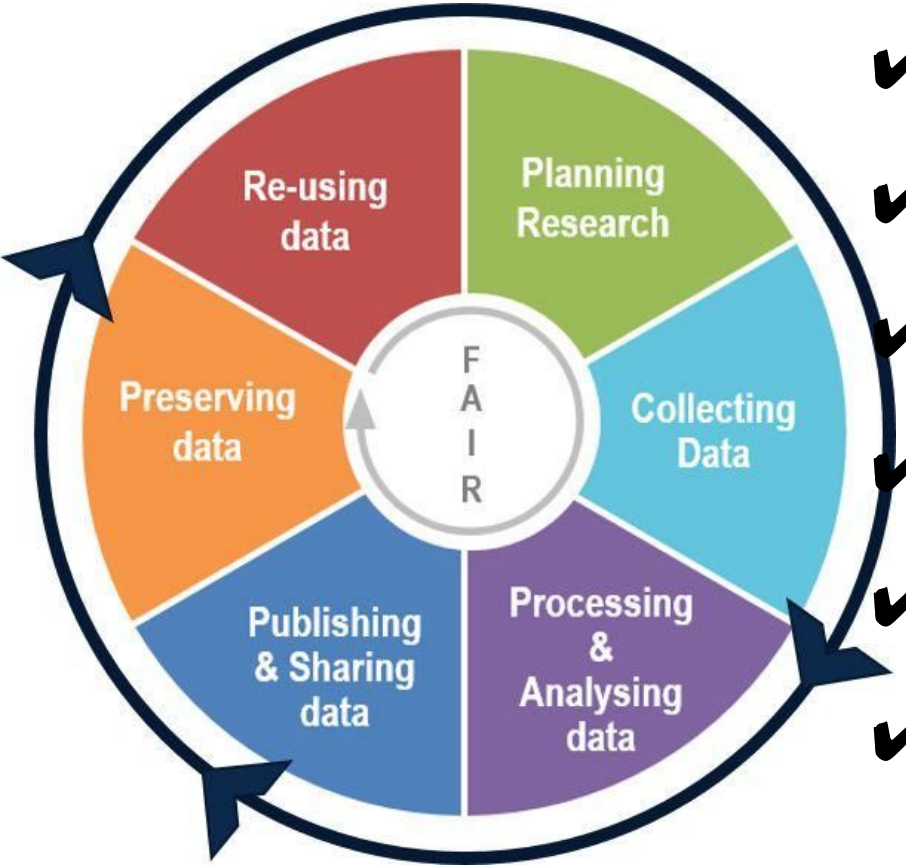


Why open Science is important?



Bertram et al., **Open science**. Current Biology, 33, R792-R797

From data generation to reuse



- ✔ Plan data management before starting research.
- ✔ Collect data with consistent metadata.
- ✔ Use documented formats and community standards.
- ✔ Store, preserve, and assign persistent identifiers.
- ✔ Share with suitable access controls and licensing.
- ✔ Enable discovery, integration, and reuse.

Why Interoperable is the bridge?

- **Technical:** compatible file formats, APIs, and machine-readable metadata.
- **Semantic:** common terminology, ontologies, and consistent definitions.
- **Syntactic:** compatible structures and data schemas.
- **Organisational:** shared policies, responsibilities, and governance.
- **Legal:** compatible licences, consent, and permitted uses.

Liepert, C., *et al.* Interoperability of digital twin data exchange in engineering processes. *Res Eng Design* **37**, 30 (2026)

A practical life-sciences example

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Imagine two laboratories studying drug resistance in cancer.

- Laboratory A reports gene expression in one format;
- Laboratory B uses another.

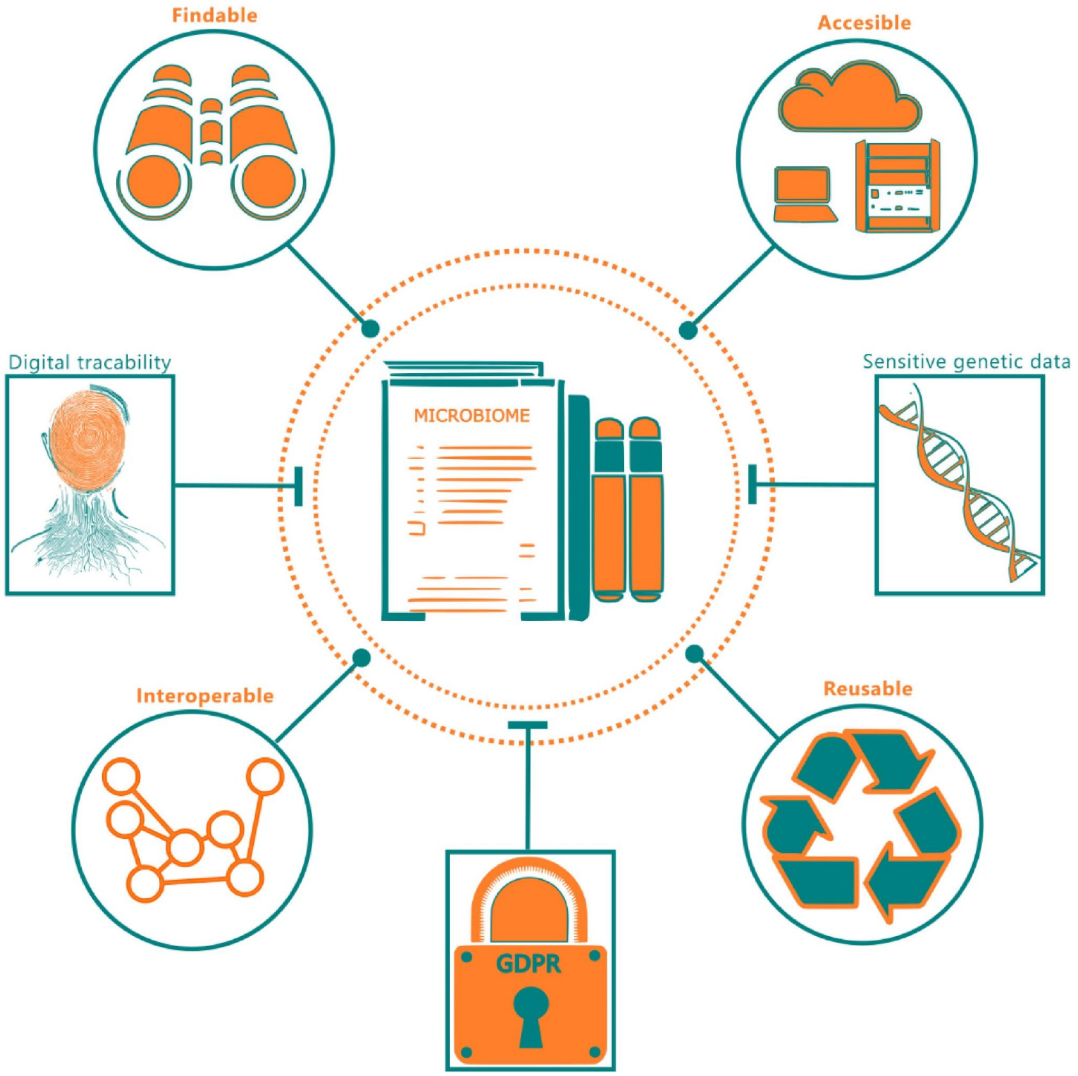
- Gene identifiers, treatment labels, units, and sample annotations differ.

- Standardised metadata, stable gene identifiers, documented protocols, and machine-readable files enable integration.

Shared analysis workflows support reproducibility and validation.

Datasets from different institutions can support a stronger, more comprehensive analysis.

Open science with responsible data governance



- privacy and confidential information
- consent, ethical approvals,
- Respect intellectual property, licences,
- Use controlled-access repositories and unrestricted sharing
- India's Digital Personal Data Protection Act, 2023.

Building institutional readiness

- Train faculty, PhD scholars, librarians, and technical staff.
- Develop institutional data management plans and repository guidance.
- Support persistent identifiers, standard metadata, and open-source tools.
- Recognize data sharing and software contributions in academic evaluation.

A roadmap for India

- **Short term: Assess institutional readiness, train researchers, and adopt minimum metadata requirements.**
- **Medium term: Improve repository interoperability, standardize identifiers and connect institutional systems.**
- **Long term: Develop a federated research data ecosystem with sustainable governance and cross-domain discovery.**

Key recommendations and conclusion

1. Make FAIR data management part of research planning.
2. Build interoperability into systems from the beginning.
3. Combine common standards with discipline-specific requirements.
4. Support openness alongside privacy, ethics, and security.
5. Invest in people, infrastructure, governance, and long-term preservation.

India's existing initiatives to highlight

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1. Indian Biological Data Centre (IBDC), New Delhi



FAIR practices:

This is an especially relevant example for genomics, proteomics, metabolomics, and structural biology

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INDA

Open-access platform for archiving and sharing nucleotide sequencing data generated in India.

Visit Portal

INDA-CA

Controlled-access platform for archiving and managing nucleotide sequencing data across India.

Visit Portal

ICPD

Open-access platform for digitization, management, storage, and exchange of crop phenotyping data.

Visit Portal

ISDA

Indian Structural Data Archive (ISDA) is a structural biology archive that contains information about the 3D shapes of proteins, nucleic acids, and complex assemblies.

Visit Portal

IMDA

IMDA is an open-access platform for archiving, managing, sharing metabolomics data and relevant information generated by MS and NMR techniques in metabolomics experiments.

Visit Portal

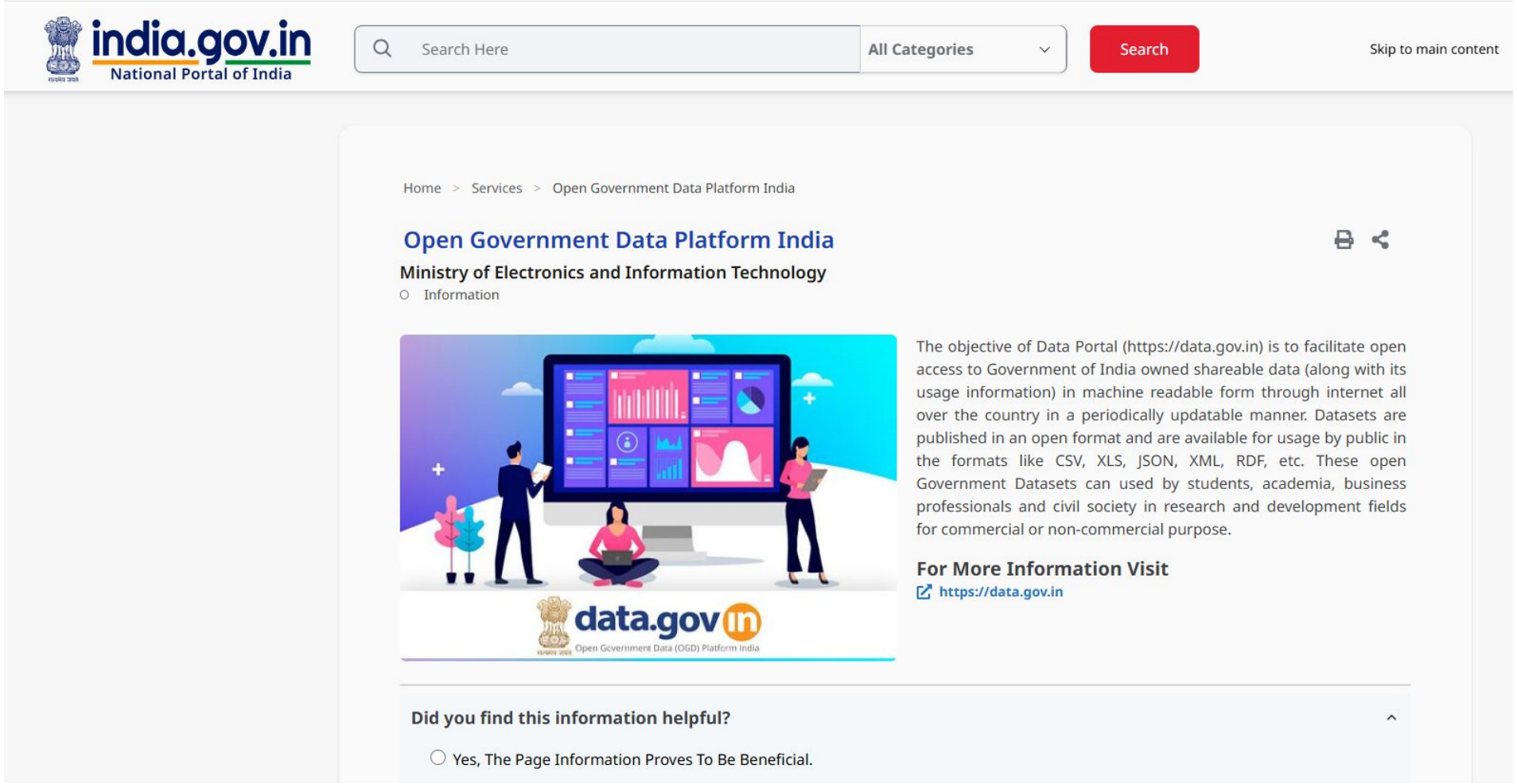
IPD

Indian Proteome Databank (IPD) is a Proteome dataset archival portal that facilitates the submission of proteome data generated by MS/MS proteomics experiments.

Visit Portal

India's existing initiatives to highlight

1. Open Government Data Platform India



India's existing initiatives to highlight

SVKM'S NMIMS Deemed to be UNIVERSITY

1. National Data Sharing and Accessibility Policy (NDSAP)



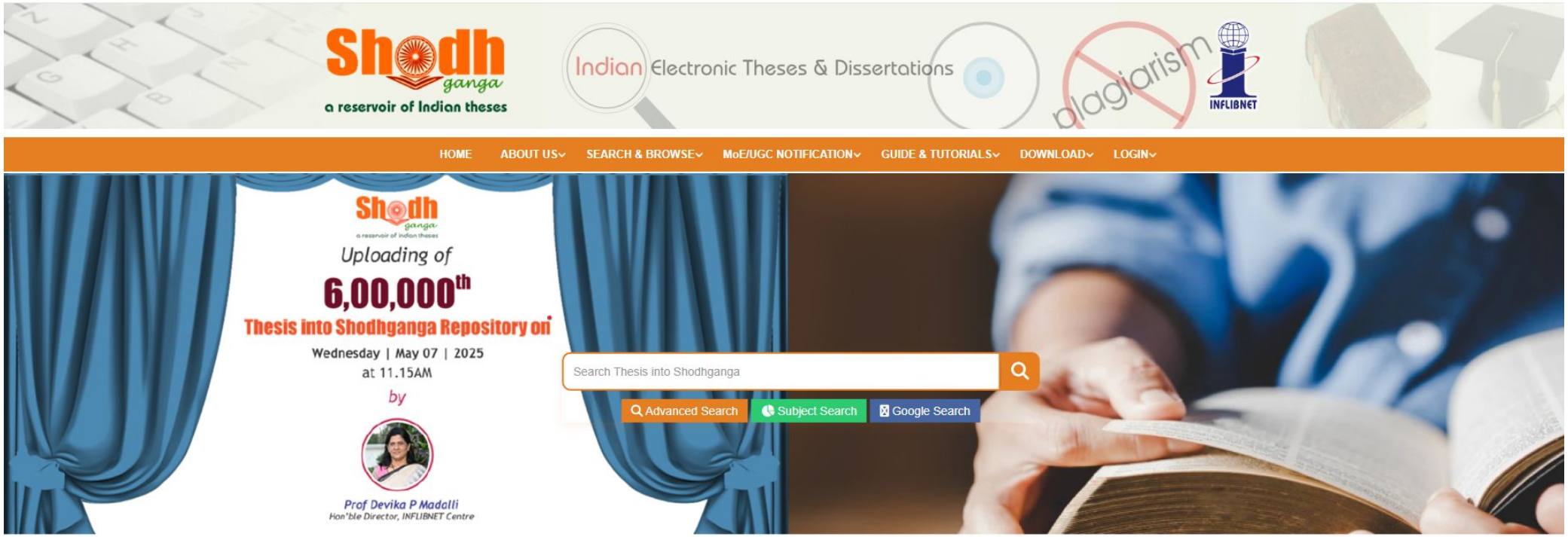
An important policy foundation for sharing non-sensitive government data generated using public funds.

It explicitly recognizes interoperability and data standards as important considerations.

India's existing initiatives to highlight

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3. Shodhganga - INFLIBNET Centre



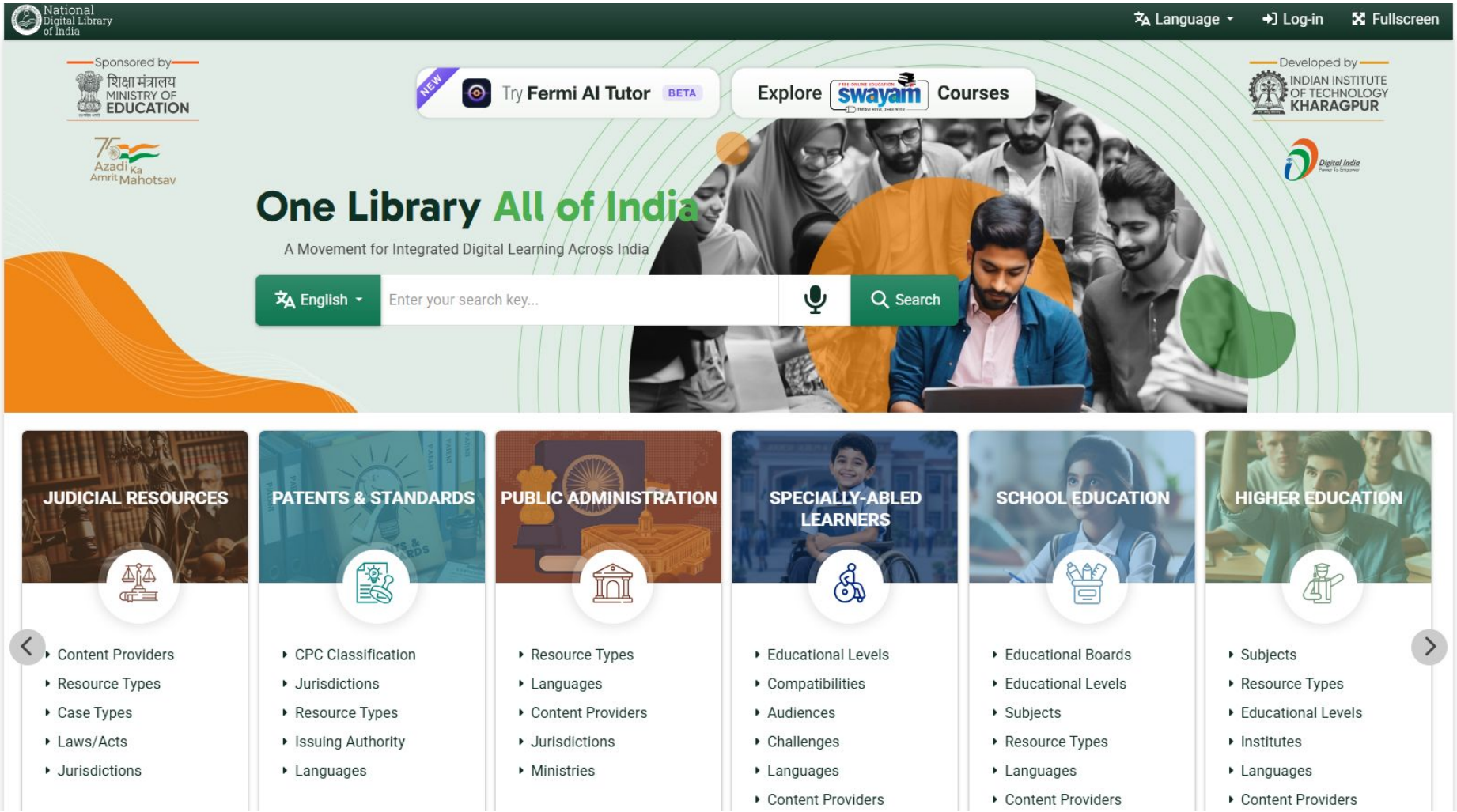
Shodhganga has crossed the remarkable miles

705196	21914	908	1060
FULL TEXT THESES	SYNOPSES/MRPs/PDFs/Fellowships	UNIVERSITIES CONTRIBUTING	Universities+CFTIs/INIs Signed MoU

India's existing initiatives to highlight

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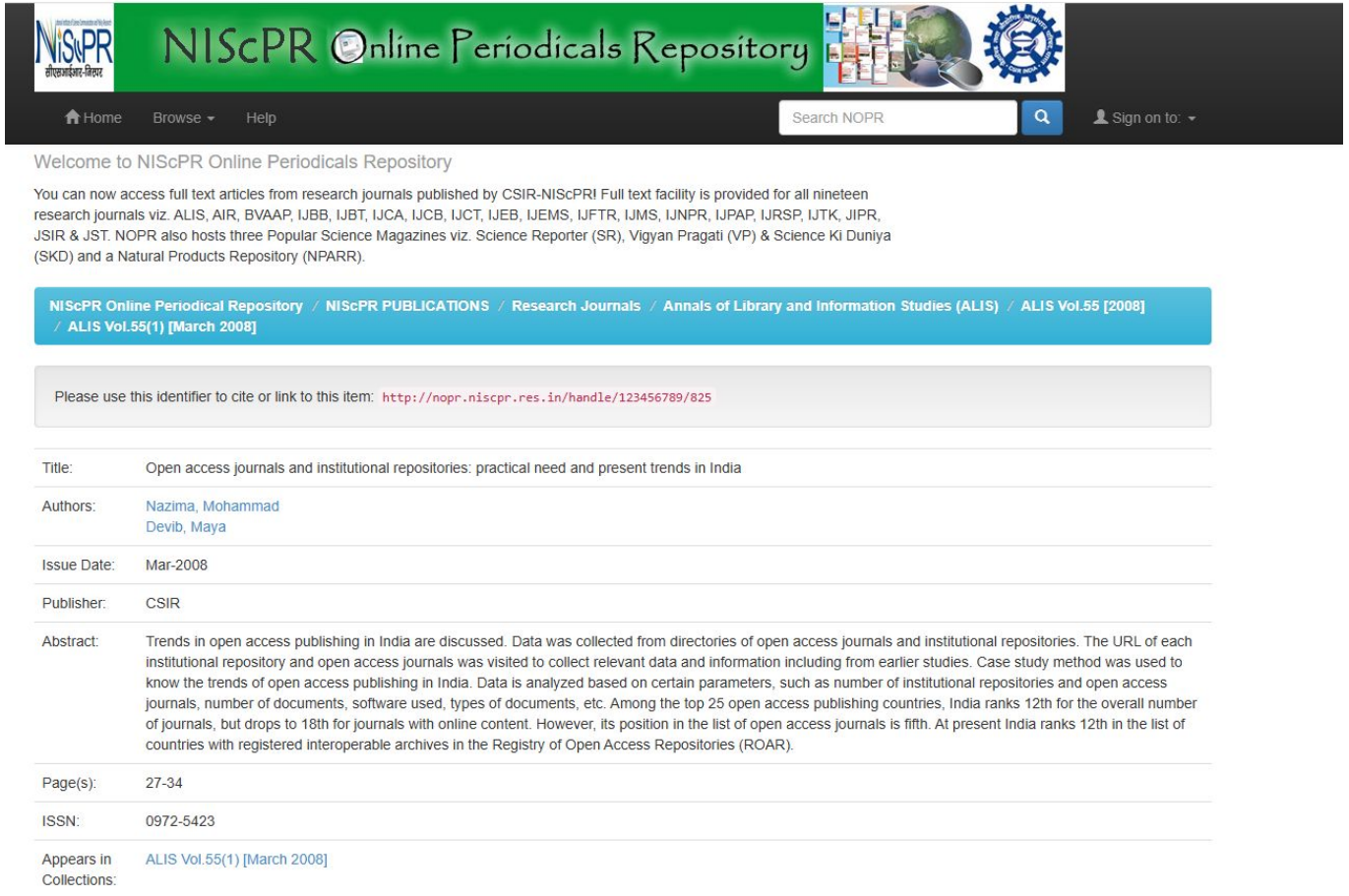
4. National Digital Library of India (NDLI)



India's existing initiatives to highlight

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5. CSIR-NIScPR Online Periodicals Repository (NOPR):

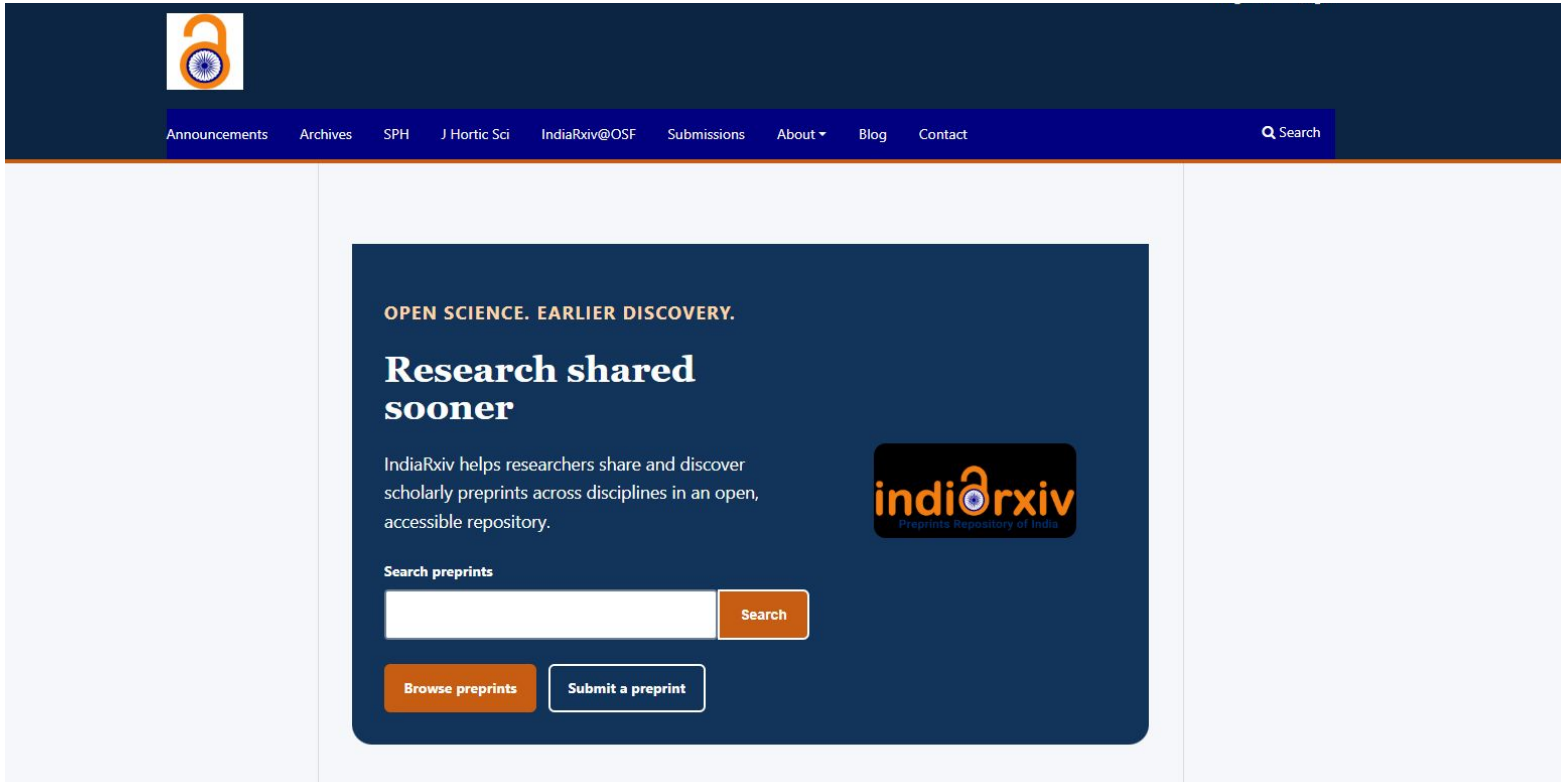


It grants completely open, full-text access to dozens of peer-reviewed scientific journals published by the institute.

India's existing initiatives to highlight

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6. IndiaRxiv:



A national preprint repository infrastructure built to host open-access, pre-publication research articles across all disciplinary sectors before they enter standard peer-review cycles

Conclusion:

India is generating an unprecedented amount of scientific data. But generating data is only the first step.

The real scientific value emerges when data can be discovered, understood, connected across disciplines, and reused to answer new questions.

This is where Open Science, FAIR principles, and interoperability come together.

India's opportunities and persistent gaps

INDIAN RESEARCH DATA ECOSYSTEM: OVERVIEW & CHALLENGES

STRUCTURAL ADVANTAGES
(ECOSYSTEM VALUE)



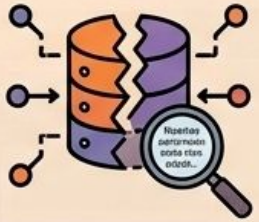
1. HIGH-VOLUME GENERATION:
Diverse research institutions generate large, valuable data streams across disciplines.





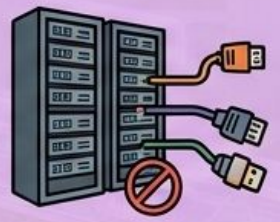
2. INTERDISCIPLINARY COLLABORATION:
Substantial opportunities for cross-institutional partnerships.

TECHNICAL BOTTLENECKS
(STRATEGIC EXECUTION GAPS)



3. UNEVEN REPOSITORY CAPABILITIES:
Repository performance varies; low metadata standard adoption.





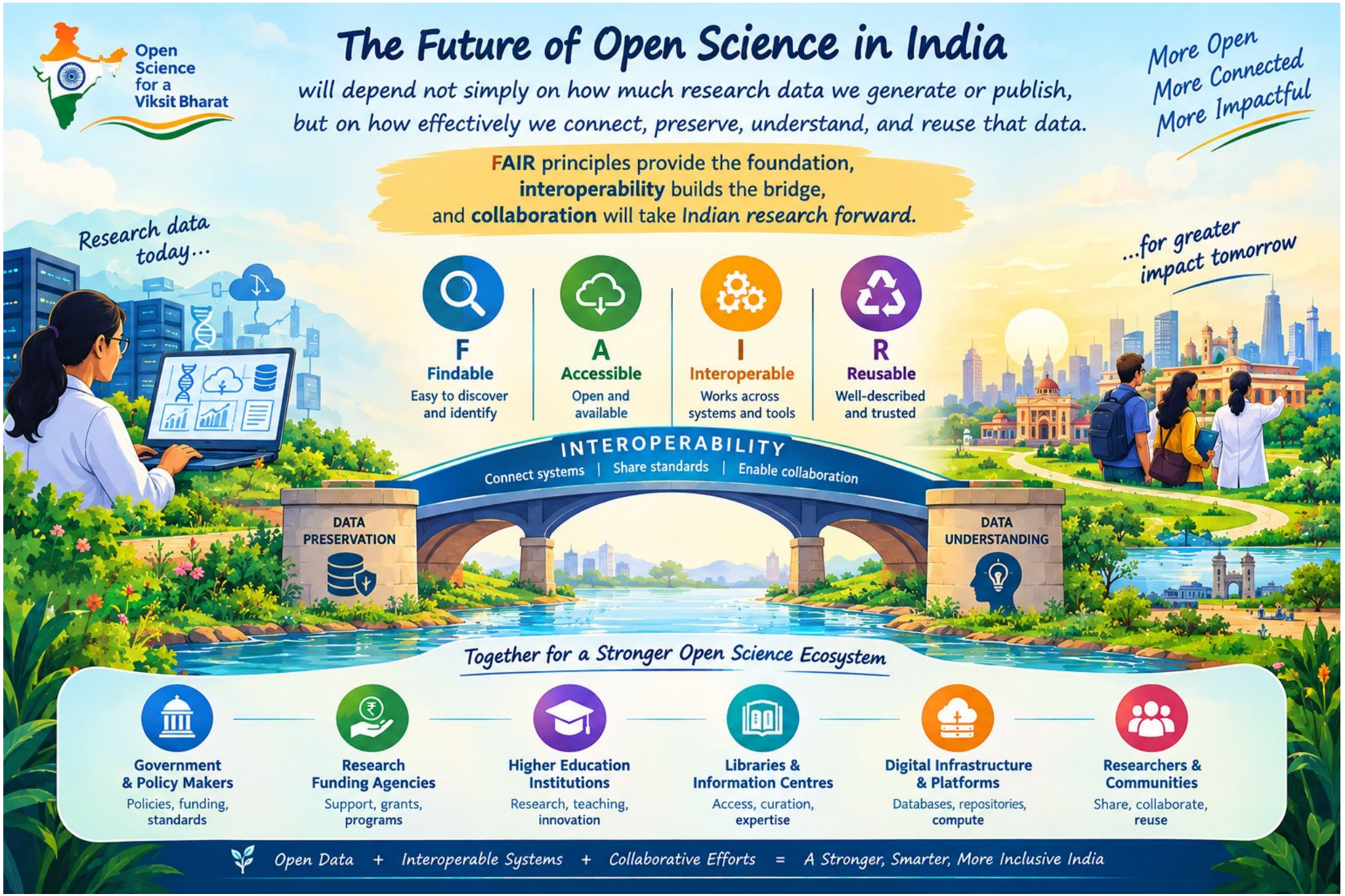
4. SYSTEM FRAGMENTATION:
Limited interoperability between institutional and external domain-specific hubs.





5. CAPACITY ENABLEMENT NEEDS:
Crucial for incentives, skill development, sustainable funding, and data stewardship.

Strategic Analysis | For Executive Discussion | [Your Organization Name/Logo]


Open Science for a Viksit Bharat
The Future of Open Science in India
will depend not simply on how much research data we generate or publish,
but on how effectively we connect, preserve, understand, and reuse that data.
More Open
More Connected
More Impactful
FAIR principles provide the foundation,
interoperability builds the bridge,
and collaboration will take Indian research forward.
Research data today...
...for greater impact tomorrow
F
Findable
Easy to discover and identify
A
Accessible
Open and available
I
Interoperable
Works across systems and tools
R
Reusable
Well-described and trusted
INTEROPERABILITY
Connect systems | Share standards | Enable collaboration
DATA PRESERVATION
DATA UNDERSTANDING
Together for a Stronger Open Science Ecosystem
Government & Policy Makers
Policies, funding, standards
Research Funding Agencies
Support, grants, programs
Higher Education Institutions
Research, teaching, innovation
Libraries & Information Centres
Access, curation, expertise
Digital Infrastructure & Platforms
Databases, repositories, compute
Researchers & Communities
Share, collaborate, reuse
Open Data + Interoperable Systems + Collaborative Efforts = A Stronger, Smarter, More Inclusive India

India's Open Science Landscape

Collaborative Knowledge | Accessible Research | A Stronger Tomorrow

Open Science builds trust, accelerates discovery and brings science closer to society.

Open Science for a Viksit Bharat

National Research Funding Agencies & Higher Education Institutions

Funding | Talent | Research Excellence

Key Funding Agencies

- **DST** (Department of Science & Technology)
- **DBT** (Department of Biotechnology)
- **ICSSR** (Social Science Research)
- **ICMR** (Medical Research)
- **CSIR** (Council of Scientific & Industrial Research)
- **UGC** (University Grants Commission)
- **SERB** (Science and Engineering Research Board)
- **MoES** (Ministry of Earth Sciences)
- **DAE** (Department of Atomic Energy)

Higher Education Institutions

- IITs
- IISc
- Central Universities
- State Universities
- Deemed-to-be Universities
- National Institutes
- Research & Innovation Hubs

Institutional Repositories, Libraries and Research Data Services

Preserve | Share | Enable Reuse

Institutional Repositories

- IITs, IISc, NITs, Central Universities
- State Universities
- Research Institutes

Libraries

- Digital & Physical Collections
- E-resources & Journals
- Research Support Services

Research Data Services

- Data Management Plans
- Data Repositories
- Metadata & DOI Assignment
- Training & Capacity Building

From information to impact

India's Open Science Ecosystem

People + Data + Infrastructure + Policy

Government-Supported Scientific Databases & Digital Infrastructures

Open Access | Interoperability | Scale

Key Scientific Databases

- **INFLIBNET** (e-Resources, Shodhganga, Shodhgangotri)
- **NCBI** (GenBank, PubMed)
- **CSIR-NIScPR** (NISCAIR, e-Books)
- **Bhuvan** (ISRO – Earth Observation)
- **IMD** (Climate & Weather Data)
- **NSO** (Statistical Data)
- **India Biodiversity Portal** (NBA)

Digital Infrastructures

- National Knowledge Network (NKN)
- e-Science Grid
- **SATHI** (National Platform for Research Data)
- **UGC CARE** (Journal & Research Evaluation)
- **DIKSHA** (Digital Learning Resources)

Data for discovery, infrastructure for innovation



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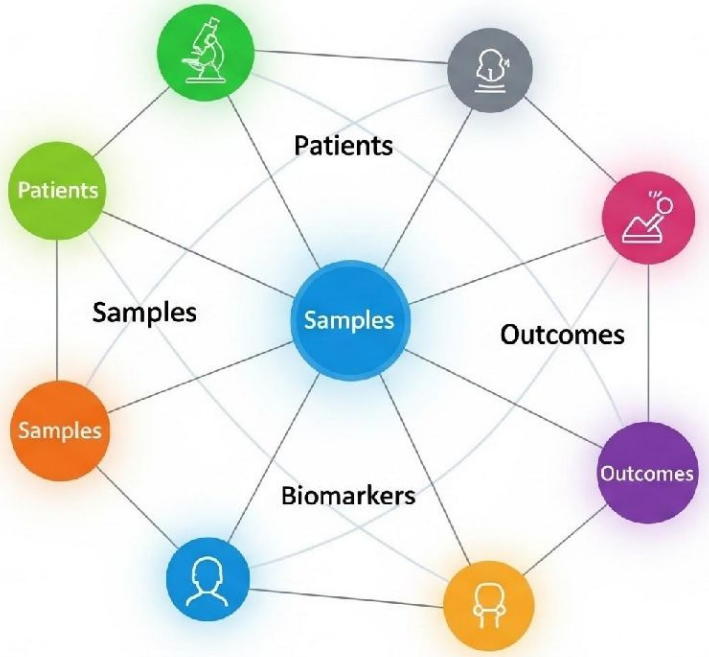
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Barriers to Open Science

Disconnected Data Silos

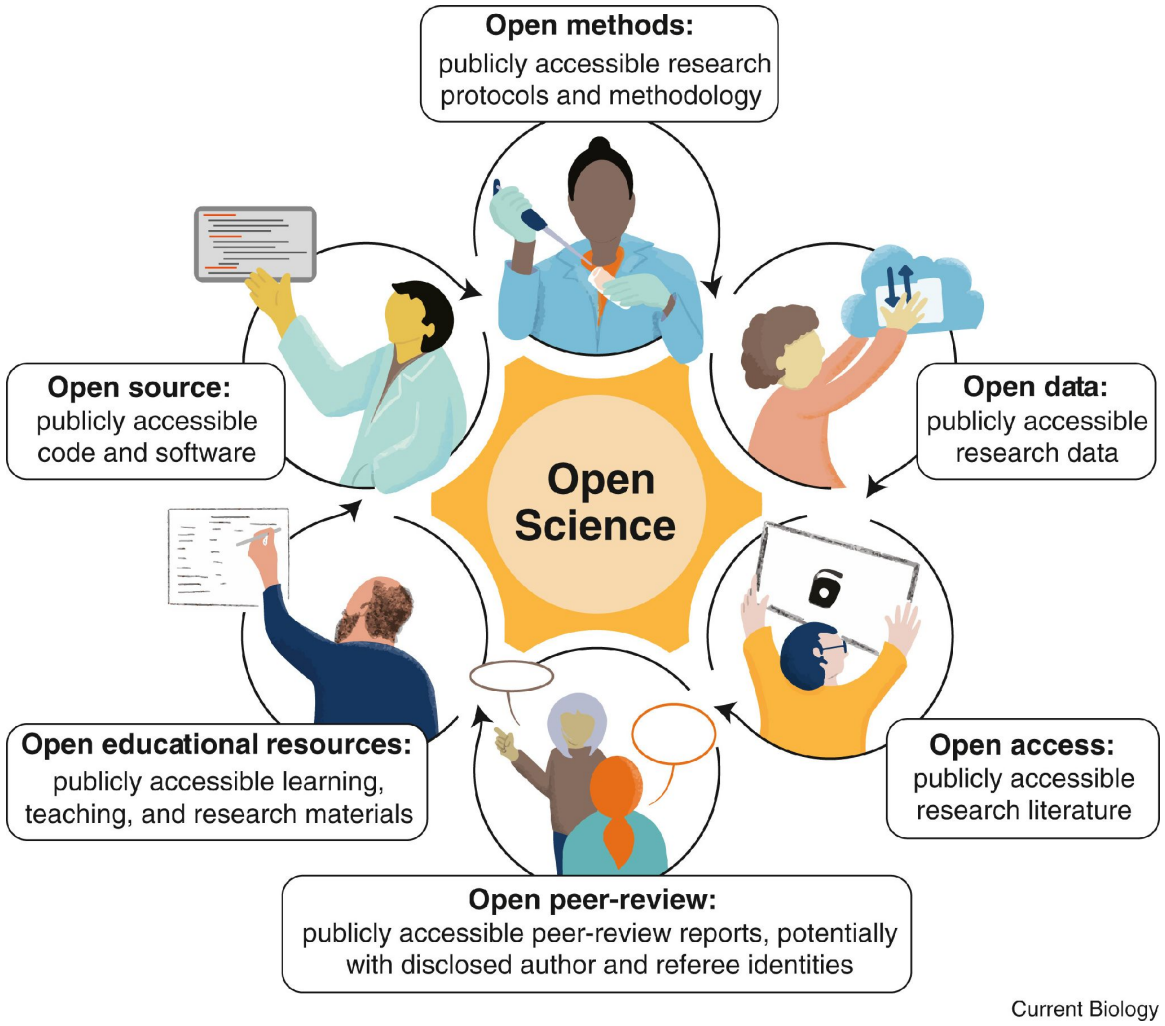


Unified Data Model



Open data alone does not guarantee reusable science.

Why open Science is important?



- Research data, publications, Tools

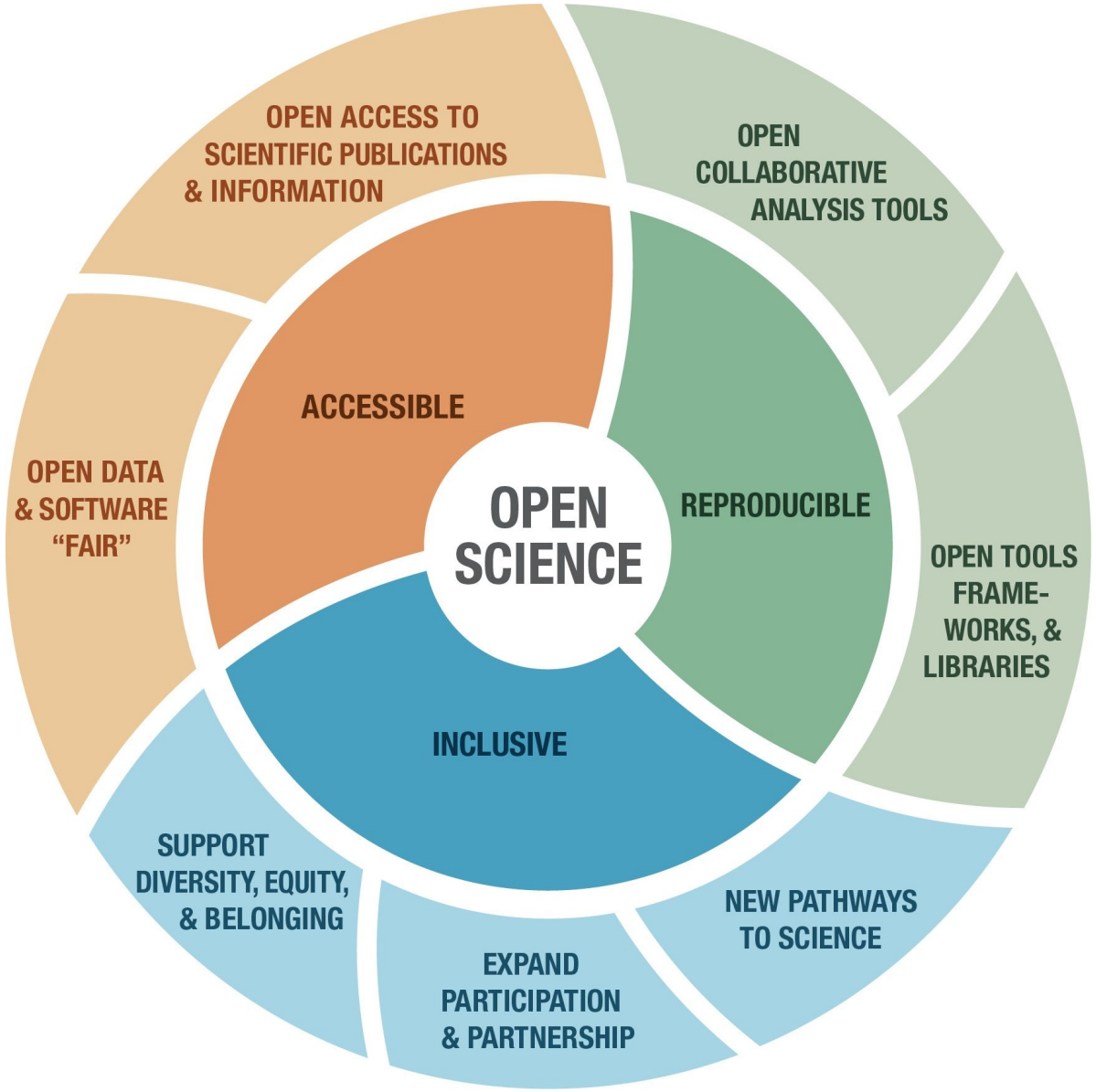
  Freely accessible

- Accelerates discovery, Builds public

  trust, and Drives global collaboration

Bertram et al., **Open science**. Current Biology, 33, R792-R797

Need of open ecosystem?



Importance of FAIR System

BEFORE



MESSY DATA

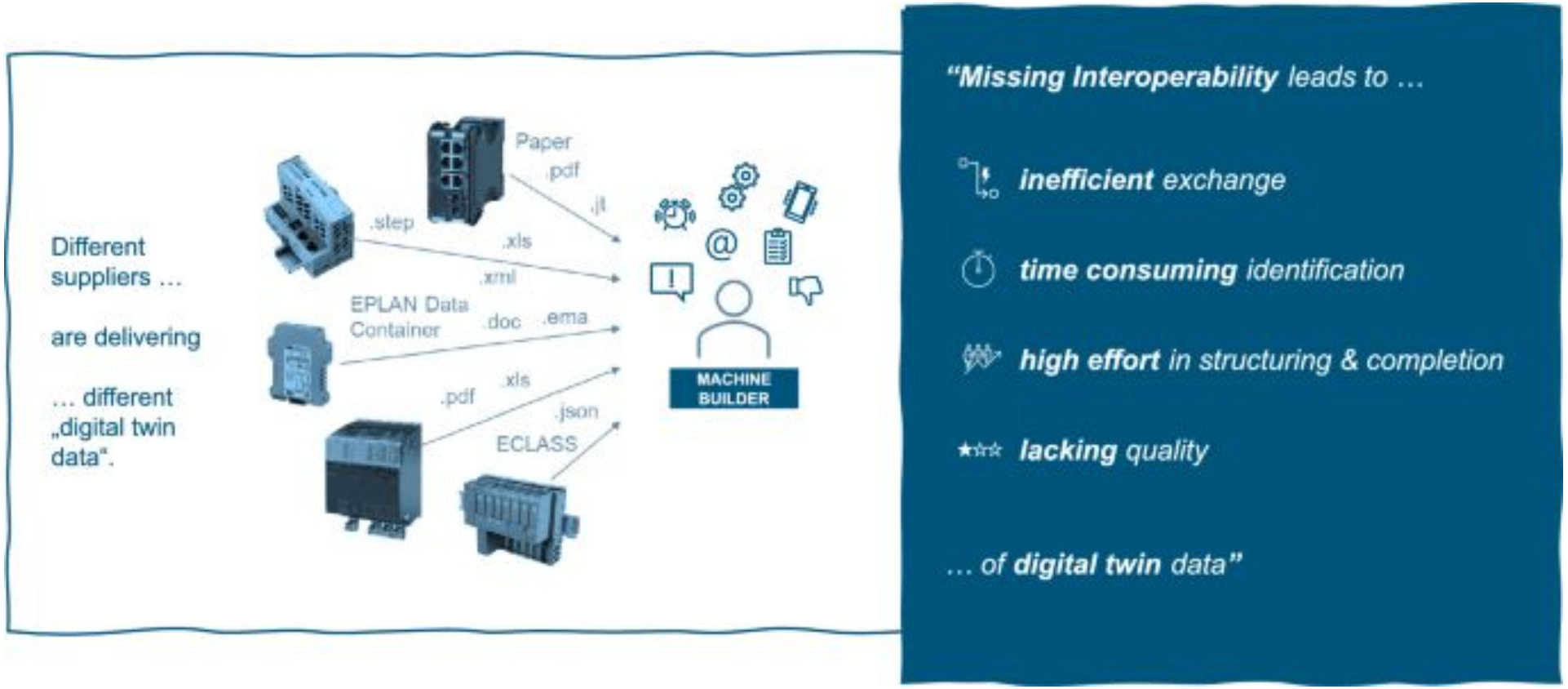
AFTER



CONNECTED, DISCOVERABLE DATA

Why Interoperable is the bridge?

The ability of different systems to work together, i.e., inter-operating, to transfer and use information in a uniform and efficient manner

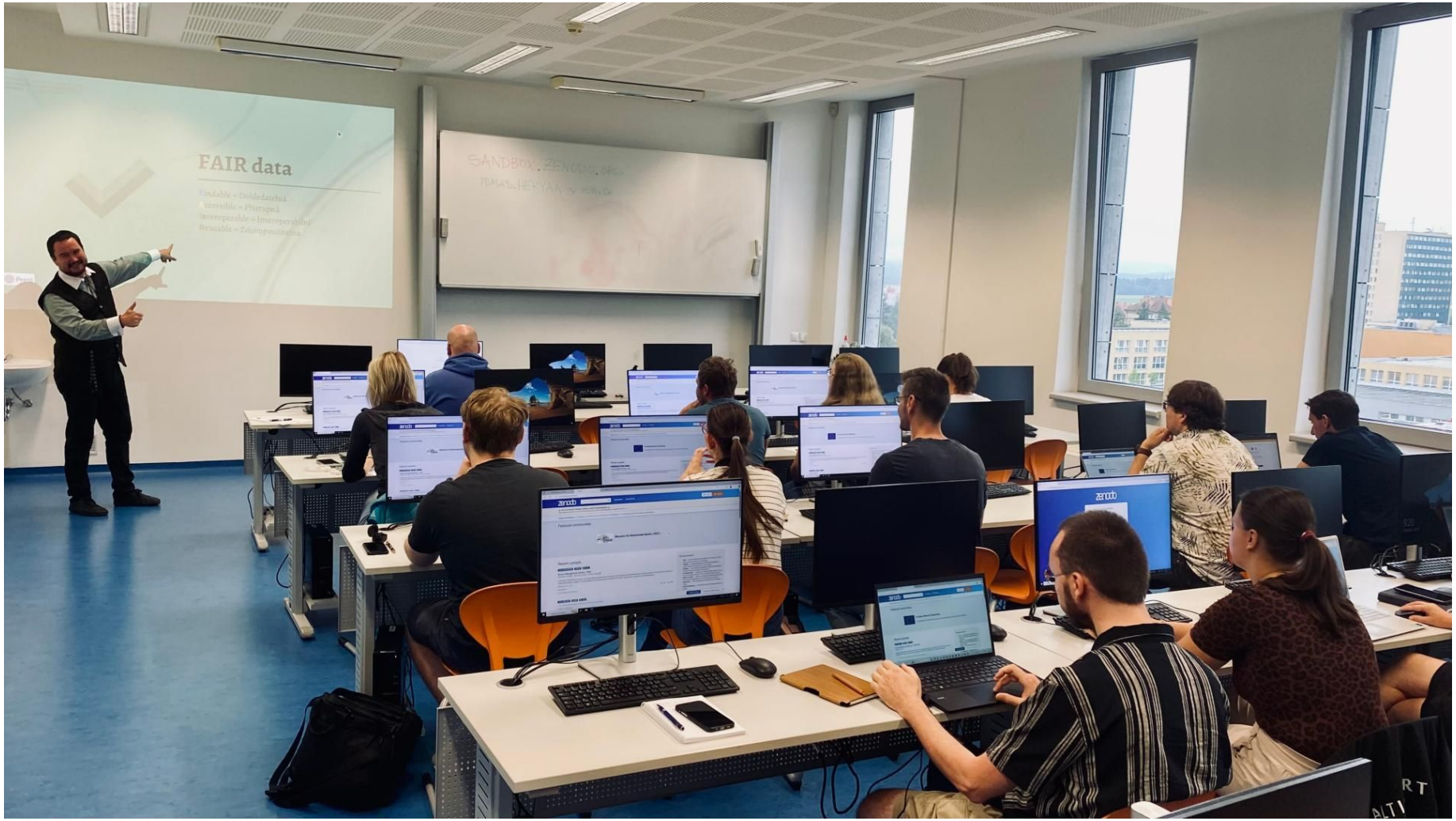


Liepert, C., *et al.* Interoperability of digital twin data exchange in engineering processes. *Res Eng Design* **37**, 30 (2026)

From data generation to reuse

- ✔ Plan data management before starting research.
- ✔ Collect data with consistent metadata.
- ✔ Use documented formats and community standards.
- ✔ Store, preserve, and assign persistent identifiers.
- ✔ Share with suitable access controls and licensing.
- ✔ Enable discovery, integration, and reuse.

Building institutional readiness



India's Open Science landscape

- National research funding agencies and higher education institutions.
- Institutional repositories, libraries, and research data services.
- Government-supported scientific databases and digital infrastructures.
- National policies on research, digital public infrastructure, and data governance.